Alperen ARSLAN

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Electrical and Electronics Engineer with hands-on experience in FPGA-based systems, laboratory-driven R&D environments, and hardware–software integration, with a focus on SoC-based architectures and timing-critical digital systems. Experience includes FPGA–SoC integration, high-resolution timestamping, secure data processing. Highly motivated to learn, continuously improve, and contribute to advanced engineering projects in multidisciplinary teams.

EXPERIENCE

Bachelor Of Science in Electrical and Electronics Engineering Sep 2019 - Jan 2024
Ozyegin University, Faculty of Engineering, Quantum – Optics Lab
Dissertation Title: Time Stamping Unit ASIC Based

QUBITRIUM Nov 2023 - Jan 2026
Digital Design Engineer – R&D Engineer

- TÜBİTAK 1501 – Satellite-Based QKD Project (KUANT-ÜS): Developed FPGA-based payload processing architectures on Artix-7, including Verilog RTL design, AXI/AXI-Lite and MicroBlaze integration, high-resolution timing and data processing, peripheral control (DDR3, ADC, SPI, UART, CAN), and system bring-up for satellite applications.
- D-Orbit & TNO – Satellite-Based QKD Projects: Designed and integrated FPGA RTL cross correlation, sycnhronizer, histogram modules with MicroBlaze systems via AXI, implemented embedded control logic and C-based software, and actively participated in system integration, on-site validation, and technical documentation.
- PURPLE NECTAR(Netherlands): Developed a C++-based timestamp data processing framework with channel-wise separation, USB communication, and optional Python API support for user-level data access and visualization.
- GUI Development – Timestamping Unit: Contributed to end-to-end GUI and firmware development, combining FPGA RTL, C#/C++ software, USB-based communication, data processing and visualization, multi-device support, and productization activities.
- Web Application Development (QKD Demo): Designed and implemented a Firebase-based video conferencing web application to demonstrate secure communication workflows within an integrated QKD system.

QUBITRIUM July 2023 – Sep 2023
Internship – R&D Engineer

- Designed and developed a DRV592-based temperature controller unit, contributing to schematic design, PCB layout, and embedded software development in collaboration with a colleague.

PROJECTS

Graduation Project - ASIC-Based Timestamping Unit Oct 2023 – Jun 2024

- Designed an ASIC-based timestamping system using TDC-GPX2, Spartan-6 FPGA, and USB 3.0, including schematic, PCB, and embedded software development.

FPGA-to-FPGA Communication (Basys3)

- Implemented SPI-based communication between two Basys3 FPGAs with UART-controlled command handling to enable data exchange between two host PCs.

AES-256 CTR Mode Hardware IP Core

- Implemented AES-256 in CTR mode using pure Verilog RTL on Zynq-7020 FPGA. 14-round pipeline with AXI-Lite control and AXI-Stream data interfaces.

Sound Recognition

- Developed a real-time sound recognition system using MATLAB HDL Coder and DSP-based signal filtering blocks, implemented on a Zynq-7020 SoC with Ethernet and audio jack interfaces, and built a C# application for system control and result monitoring.

Wireless Virtual Display over FPGA-based HDMI Output

- Designed and implemented an end-to-end system that captures a Windows virtual on a host PC and streams it in real time to a FPGA development board for HDMI output at 1080p@60Hz.

SKILLS

Computer Skills: Vivado, Vitis, Visual Studio, STM32CubeIde, Altium Designer, MATLAB, Kicad, LTSpice, Git, Fork, Office Apps
Programming Languages: Verilog, C, C++, C#, Python, Javascript
Languages: English (Fluent), Turkish (Native)

ATTENDED EVENTS

As Exhibitor

SAHA EXPO - International Defense, Aerospace, and Space Industry Fair (Istanbul) Oct 2024
- Responsible for product presentation and project introduction to visitors and industry stakeholders.

QEXPO 2025 (International Exhibition on Quantum Technologies) May 2024
- Responsible for product presentation and project introduction to visitors and stakeholders.

NATO's Innovation Continuum Oct 2025
- Actively involved in the preparation and live execution of the project demonstration. (link: https://www.act.nato.int/article/shine-2025/)

ADDITIONAL INFORMATION

- Ozyegin University Bicycle board member - 2023
- Ozyegin University SAS board member - CMAS 1 Star diver2024
- Driving License B1
- Drone Driver License
- Completion of military service